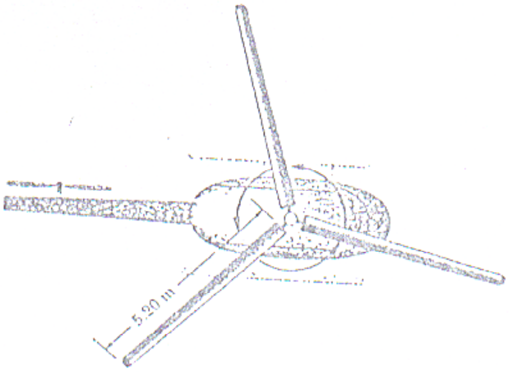
(g) Elkeen van die drie lemme van 'n helikopter se rotor het 'n massa van 240 kg en is 5,2 m lank. Toon aan dat die traagheidsmoment van die rotor is $I = 6{,}49 \times 10^3$ kg m^2.
Each of the three helicopter rotor blades is 5,2 m long and has a mass of 240 kg. Show that the rotational inertia of the rotor is $I = 6{,}49 \times 10^3$ kg m^2.



$I = I_1 + I_2 + I_3 = 3 I_{(\text{single blade})} = 3\left(\frac{1}{3}(240)(5{,}2)^2\right)$
$= 6489{,}6 \text{ kg.m}^2$
$= 6{,}49 \times 10^3 \text{ kg.m}^2$

3

(3)

(h) Indien die rotor draai teen 350 rev/min, bepaal die rotasie kinetiese energie.
If the rotor is rotating at 350 rev/min, what is the kinetic energy of rotation.

$\omega = 350$ rpm $= 2\pi \times 350$ rad $= 36{,}65$

$K = \frac{1}{2} I \omega^2 = \frac{1}{2}(6{,}49 \times 10^3)(36{,}65)^2$
$= 4358757$ J
$= 4{,}36 \times 10^6$ J

4

(4)

(i) 'n Ingenieur neem 'n paar van sy vriende met sy helikopter vir 'n naweek na sy jagplaas. Toe hulle moet terugkeer wil die helikopter nie aanskakel nie en hy bedink 'n plan om met toue die drie lemme te draai. Persoon A trek met 'n krag $F_A = 50$ N, B met $F_B = 60$ N en C met $F_c = 70$ N. Al drie die toue maak 'n hoek van 30° met die horisontaal. A en B se toue is loodreg op die lemme, maar C se tou maak 'n hoek van 10°, soos in die skets.
An engineer takes a few friends to his farm for a weekend with his helicopter. When they decided to return home, the helicopter did not start. He then got a plan to rotate the engine with three ropes tied to the blades. Person A pulls with $F_A = 50$ N, *B with* $F_B = 60$ N *and C with* $F_c = 70$ N. *The three ropes each has an angle of 30° relative to the horizontal. The ropes of A and B are perdendicular to the blades, but rope C is at an angle of 10° relative to the blade – as shown in the figure.*

Vraag / *Question* 3(i)(Vervolg / *Continue*)/ ....